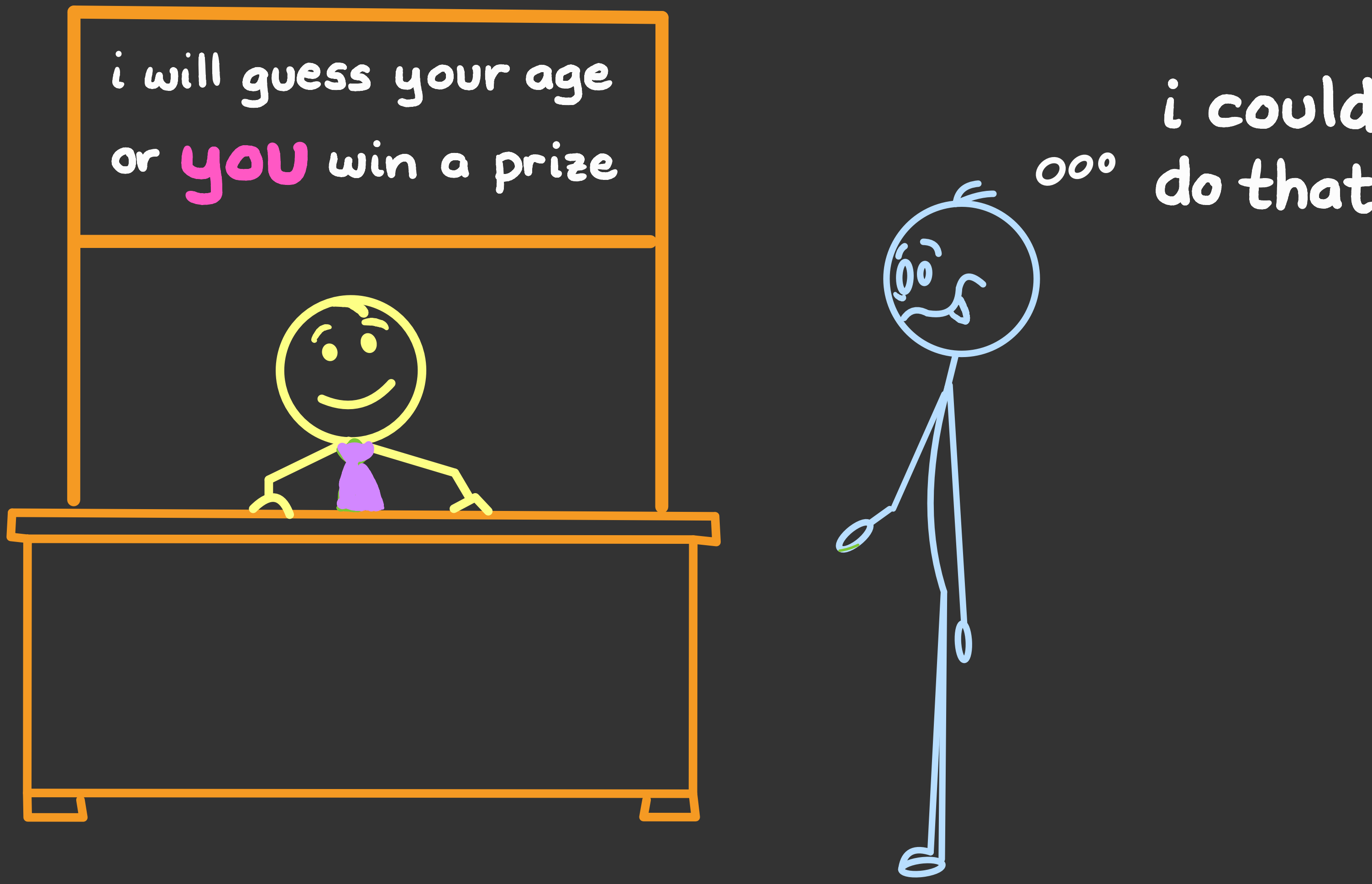
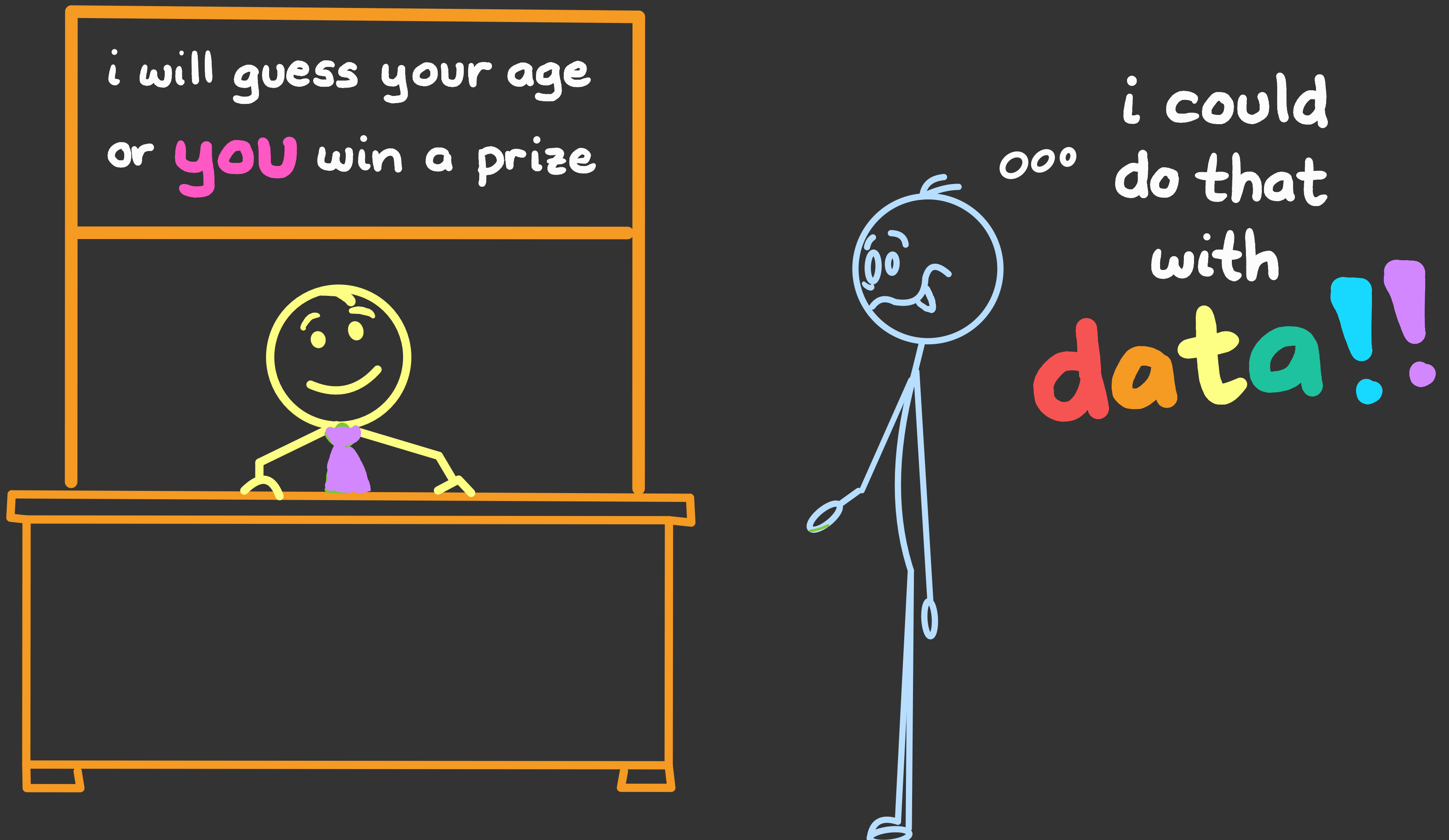
machine learning

CSCI 270

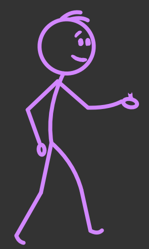

i will guess your age
or you win a prize
i could do that


i will guess your age
or you win a prize
ooo
i could do that with data!!

your friends

alice
age 20
has taken 5 cs courses

bob
age 28
has taken 8 cs courses



chandra
age 41
has taken 12 cs courses

hypothesis

a person's age is a multiple of the number of cs courses they've taken

$$y = \theta x$$

age

num cs courses

unknown multiplier

we want to choose θ such that

$$20 \approx 5\theta$$

$$28 \approx 8\theta$$

$$41 \approx 12\theta$$

$$y = \theta x$$

age

unknown multiplier

num cs courses

your friends

	alice	bob	chandra
age	20	28	41
cs courses	5	8	12

we want to choose θ such that

$$20 \approx 5\theta$$

$$28 \approx 8\theta$$

$$41 \approx 12\theta$$

how do we express **this** as a mathematical optimization?

we want to choose θ such that

$$20 \approx 5\theta$$
$$28 \approx 8\theta$$
$$41 \approx 12\theta$$

find θ that minimizes

$$|20 - 5\theta| + |28 - 8\theta| + |41 - 12\theta|$$

we want to choose θ such that

$$20 \approx 5\theta$$
$$28 \approx 8\theta$$
$$41 \approx 12\theta$$

or: find θ that minimizes

$$(20 - 5\theta)^2 + (28 - 8\theta)^2 + (41 - 12\theta)^2$$

find θ that minimizes

$$|20 - 5\theta| + |28 - 8\theta| + |41 - 12\theta|$$

$$\boxed{\underset{\theta}{\operatorname{argmin}} \; |20 - 5\theta| + |28 - 8\theta| + |41 - 12\theta|}$$

or: find θ that minimizes

$$(20-5\theta)^2+(28-8\theta)^2+(41-12\theta)^2$$

$$\boxed{\underset{\theta}{\operatorname{argmin}}\ (20-5\theta)^2+(28-8\theta)^2+(41-12\theta)^2}$$

$$L_1(\theta) = |20 - 5\theta| + |28 - 8\theta| + |41 - 12\theta|$$

or: $$L_2(\theta) = (20 - 5\theta)^2 + (28 - 8\theta)^2 + (41 - 12\theta)^2$$

want to compute: $\underset{\theta}{\operatorname{argmin}} \; L(\theta)$

where $L \in \{L_1, L_2\}$

these are called loss functions

they quantify the amount of embarrassment we feel
if we use a particular value of θ to make predictions

$$L_1(\theta) = |20 - 5\theta| + |28 - 8\theta| + |41 - 12\theta|$$

$$L_2(\theta) = (20 - 5\theta)^2 + (28 - 8\theta)^2 + (41 - 12\theta)^2$$

suppose we use $\theta = 10.0$

$$y = \theta x$$

age

unknown multiplier

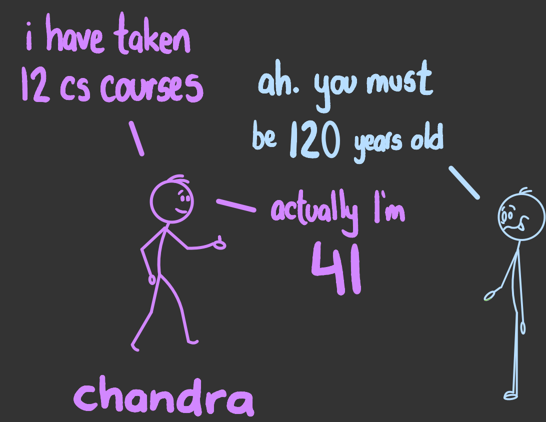
num cs courses

according to L_1 loss,

this produces

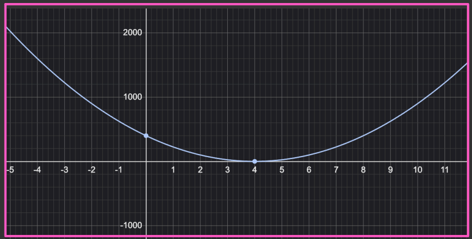
$$120 - 41 = 79$$

units of embarrassment 😅

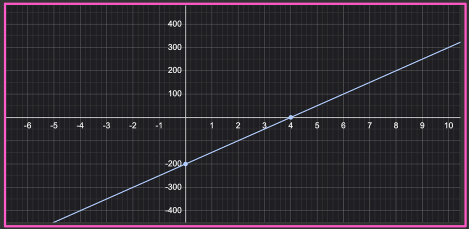


let's minimize our embarrassment using calculus

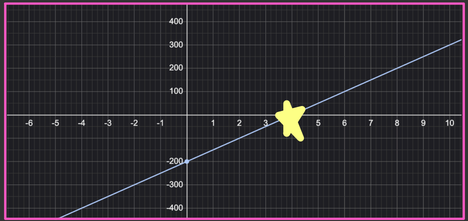
- given loss function $L(\theta)$



- compute $\dfrac{dL}{d\theta}$



- set $\dfrac{dL}{d\theta} = 0$ and solve for θ



let's minimize our embarrassment using calculus

$$\frac{dL_2}{d\theta}$$

$$= \frac{d}{d\theta}(20 - 5\theta)^2 + \frac{d}{d\theta}(28 - 8\theta)^2 + \frac{d}{d\theta}(41 - 12\theta)^2$$

$$= 2(20 - 5\theta)(-5) + 2(28 - 8\theta)(-8) + 2(41 - 12\theta)(-12)$$

$$= (-200 + 50\theta) + (-448 + 128\theta) + (-984 + 288\theta)$$

$$= 466\theta - 1632$$

let's minimize our embarrassment using calculus

$$\text{since} \quad \frac{dL_2}{d\theta} = 466\theta - 1632$$

$$\frac{dL_2}{d\theta} = 0 \quad \Rightarrow \quad 466\theta - 1632 = 0$$

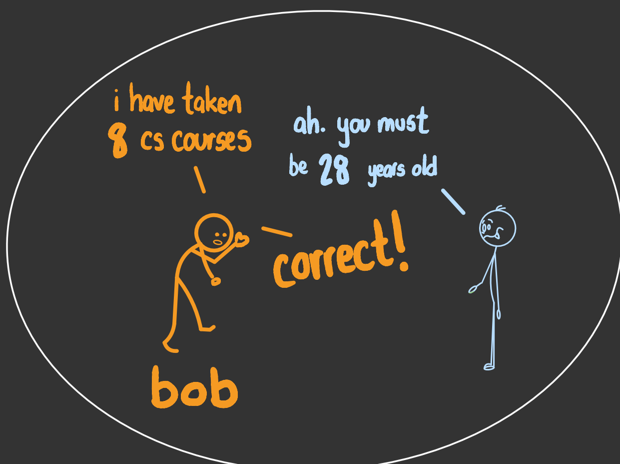
$$\Rightarrow \quad 466\theta = 1632$$

$$\Rightarrow \quad \theta \approx 3.5$$

suppose we use

$\theta = 3.5$

to make predictions

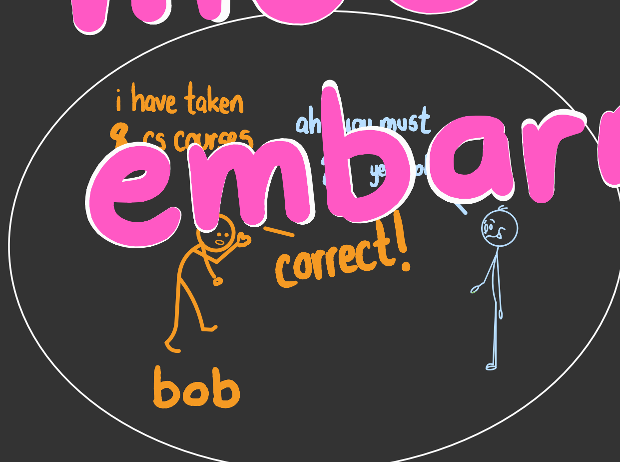


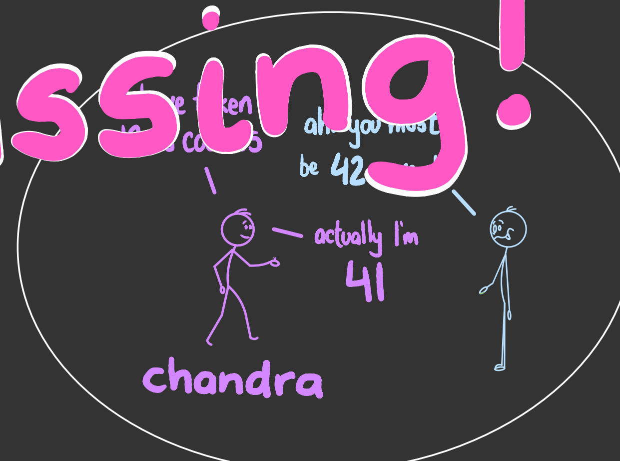
suppose we use

$\theta = 3.5$

to make predictions

much less embarrassing!

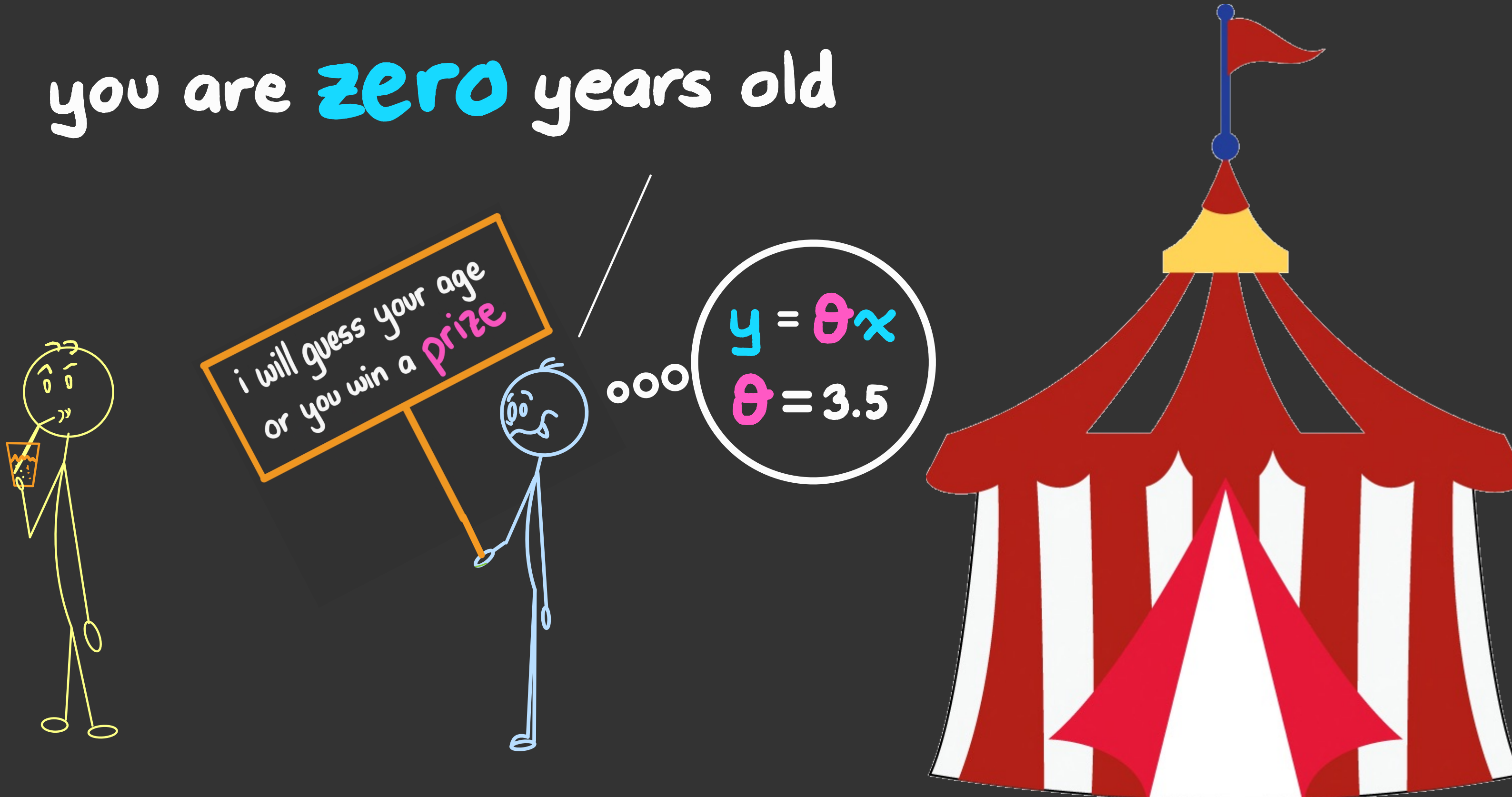




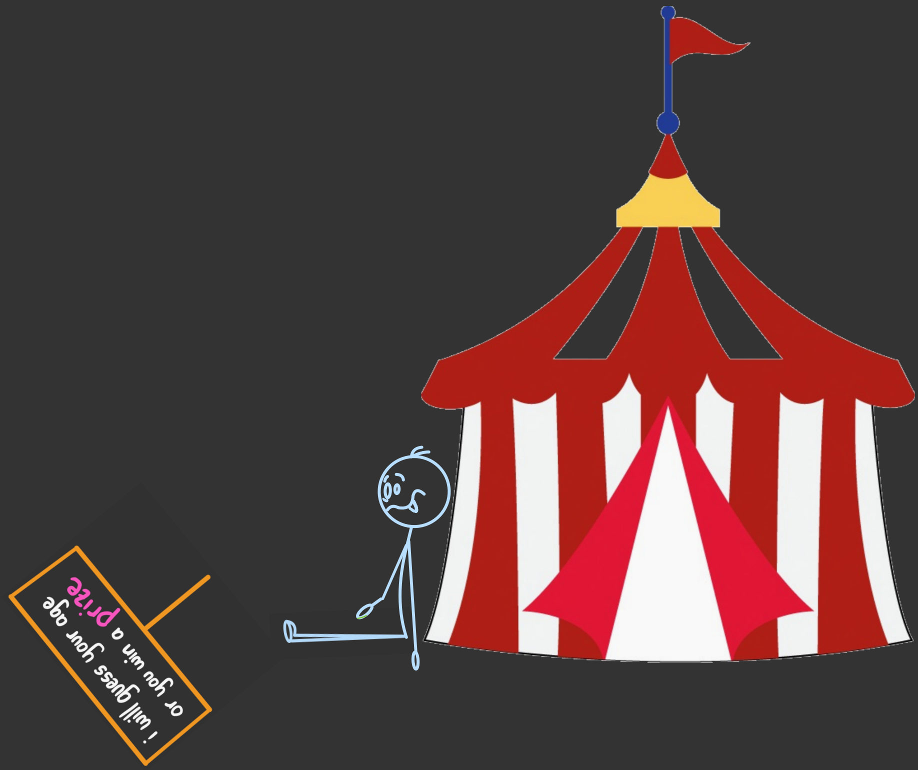
epilogue

you take your show on the road

you are **zero** years old



unfortunately the average carnival attendee
has taken **zero** computer science classes



you quickly go out of business

but what did **you** learn ?

the machine learning process

model space

$$y = \theta x$$

loss function

$$L(\theta) = (20 - 5\theta)^2 + (28 - 8\theta)^2 + (41 - 12\theta)^2$$

optimization

$$\underset{\theta}{\operatorname{argmin}}\ L(\theta)$$

training data

alice
age 20
cs courses 5

bob
age 28
cs courses 8

chandra
age 41
cs courses 12

test data

makes sure we don't overfit the training data